Experiment 1 : PCA on the Wine Dataset

Aim: Apply PCA for dimensionality reduction on the wine dataset and inspect whether the wine g
separate using a small number of principal components.

In [1]:
```
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import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
```

In [2]:
```
url = "https://media.geeksforgeeks.org/wp-content/uploads/Wine.csv"
df = pd.read_csv(url)
print("Shape:", df.shape)
```

```
Shape: (178, 14)
```

In [3]:
```
print(df.head())
```

```
   Alcohol  Malic_Acid   Ash  Ash_Alcanity  Magnesium  Total_Phenols  \
0    14.23        1.71  2.43          15.6        127           2.80
1    13.20        1.78  2.14          11.2        100           2.65
2    13.16        2.36  2.67          18.6        101           2.80
3    14.37        1.95  2.50          16.8        113           3.85
4    13.24        2.59  2.87          21.0        118           2.80

   Flavanoids  Nonflavanoid_Phenols  Proanthocyanins  Color_Intensity   Hue
0        3.06                  0.28             2.29             5.64  1.04
1        2.76                  0.26             1.28             4.38  1.05
2        3.24                  0.30             2.81             5.68  1.03
3        3.49                  0.24             2.18             7.80  0.86
4        2.69                  0.39             1.82             4.32  1.04

   OD280  Proline  Customer_Segment
0   3.92     1065                 1
1   3.40     1050                 1
2   3.17     1185                 1
3   3.45     1480                 1
4   2.93      735                 1
```

In [4]:
```
print("Columns:", list(df.columns))
```

```
Columns: ['Alcohol', 'Malic_Acid', 'Ash', 'Ash_Alcanity', 'Magnesium',
'Total_Phenols', 'Flavanoids', 'Nonflavanoid_Phenols', 'Proanthocyanins',
'Color_Intensity', 'Hue', 'OD280', 'Proline', 'Customer_Segment']
```

In [5]:
```
X = df.drop(columns=["Customer_Segment"]).values
y = df["Customer_Segment"].values
print("Features:", X.shape, "| Labels:", y.shape)
```

```
Features: (178, 13) | Labels: (178,)
```

In [6]:
```
print("Segment counts:")
print(pd.Series(y).value_counts())
```

```
Segment counts:
2    71
1    59
3    48
Name: count, dtype: int64
```

In [7]:
```
scaler = StandardScaler()
X = scaler.fit_transform(X)
print("First row after scaling:", X[0].round(2))
```

```
First row after scaling: [ 1.52 -0.56  0.23 -1.17  1.91  0.81  1.03 -0.66  1.
0.25  0.36  1.85
  1.01]
```

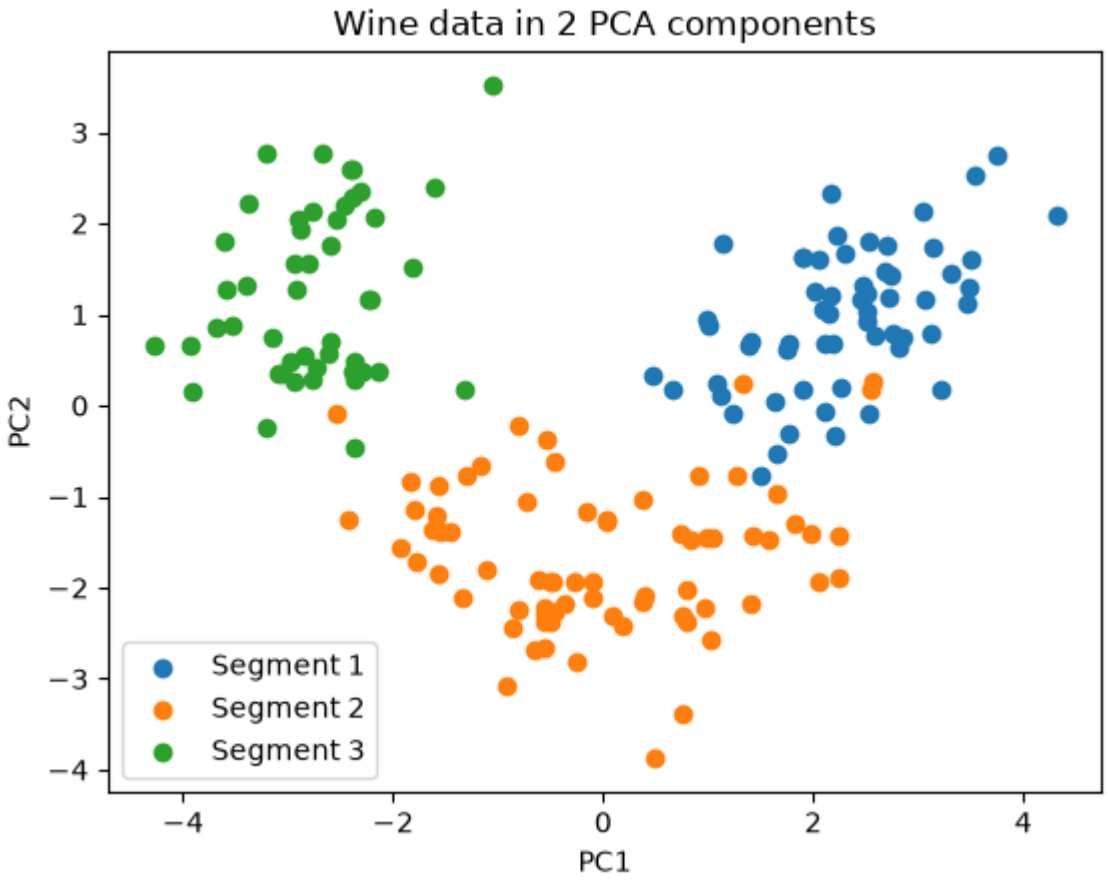
In [8]:
```
pca = PCA(n_components=2)
X2 = pca.fit_transform(X)
print("Shape after PCA:", X2.shape)
```

```
Shape after PCA: (178, 2)
```

In [9]:
```
print("Variance kept by 2 components:", round(pca.explained_variance_rati
print("Variance per component:", [round(v, 4) for v in pca.explained_vari
```

```
Variance kept by 2 components: 0.5541
Variance per component: [np.float64(0.362), np.float64(0.1921)]
```

In [10]:
```
for segment in sorted(set(y)):
    pts = X2[y == segment]
    plt.scatter(pts[:, 0], pts[:, 1], label=f"Segment {segment}")
plt.xlabel("PC1")
plt.ylabel("PC2")
plt.title("Wine data in 2 PCA components")
plt.legend()
plt.show()
```


Wine data in 2 PCA components
PC2
PC1
Segment 1
Segment 2
Segment 3
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4